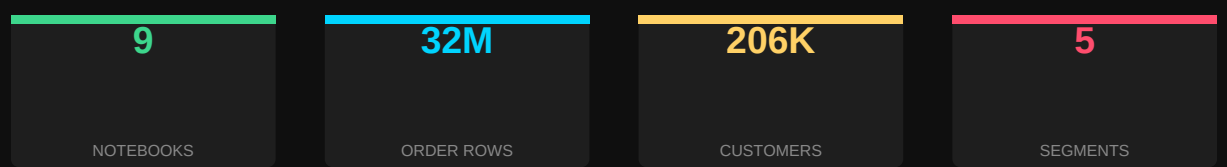


CASE STUDY

Instacart Market Basket Analysis

Customer segmentation and behavioral profiling across ~32 million order rows using Python, Pandas, and Matplotlib.



Python	Pandas	NumPy	Matplotlib	Seaborn	Jupyter
Author	Ageel Alramadhan		Program	CareerFoundry — Data Analytics	
Year	2025		Certification	DEKRA-certified · AfA-approved · 1221 UE	
Dataset	Instacart Market Basket (orders, products, customers)				
Scale	~32M order rows · ~206K customers · 49 departments				

Background & Business Questions

Instacart needed insight into customer purchasing behavior to improve targeting and marketing. This analysis used order-level data merged with customer demographics to answer four core business questions.

Business Questions

#	Business Question	Analysis Approach
1	When are customers most active? Which times/days drive orders?	Order timing analysis + busiest period flags
2	What distinguishes loyal customers from new ones?	Loyalty flag based on order count thresholds
3	How does demographics relate to spending behavior?	Age, income, and regional segmentation
4	Which customer profiles are highest value?	Multi-variable profiling (age × income × frequency)

Pipeline Architecture

Step	Notebook	Action
1. Import	4.2 IC Data import	Import orders.csv + products.csv, profile data
2. Wrangling	4.3 IC Wrangling	Rename columns, create subsets, export cleaned data
3. Consistency	4.4 IC Consistency	Fix mixed types, handle 260K missing values
4. Merging	4.5 IC Combining	Merge orders + order_products + products
5. Derived Vars	4.6 IC Deriving	Create price_label, busiest_day, busiest_period_of_day
6. Grouping	4.7 IC Grouping	Loyalty flag, spending flag, order frequency flag
7. Customers	4.8 IC Wrangling	Merge customer demographics (206K customers)
8. Viz	4.8 IC Visualizations	8 charts: orders, loyalty, prices, demographics
9. Profiling	4.9 IC Final Analysis	PII removal, regional segmentation, profiles

Datasets, Quality Checks & Preparation

Datasets Used

Dataset	Rows	Key Columns	Notes
orders.csv	~3.4M	order_id, user_id, order_hour_of_day, days_since_prior_order	Excludes eval_set on import
products.csv	49,688	product_id, product_name, aisle_id, department_id	Price anomaly: max=99,999
order_products_prior.csv	~32M	order_id, product_id, reordered	Largest file — drives order-level analysis
customers.csv	~206K	user_id, age, income, state, gender, num_dependants	Merged after order analysis

Data Quality Issues Found & Resolved

Issue	Column	Resolution
Price outlier (max = 99,999)	products.prices	Data entry error — labeled High in price_label
260,290 missing values	days_since_prior_order	First-ever orders have no prior order — filled with 0
Mixed data type (object)	days_since_prior_order	Converted with pd.to_numeric(errors="coerce")
Unintuitive column name	order_dow	Renamed to order_day_of_week
PII in customer data	first_name, last_name	Dropped before profiling and reporting
Duplicate total_orders columns	total_orders (_x/_y)	Rebuilt cleanly with groupby + merge

KEY INSIGHT

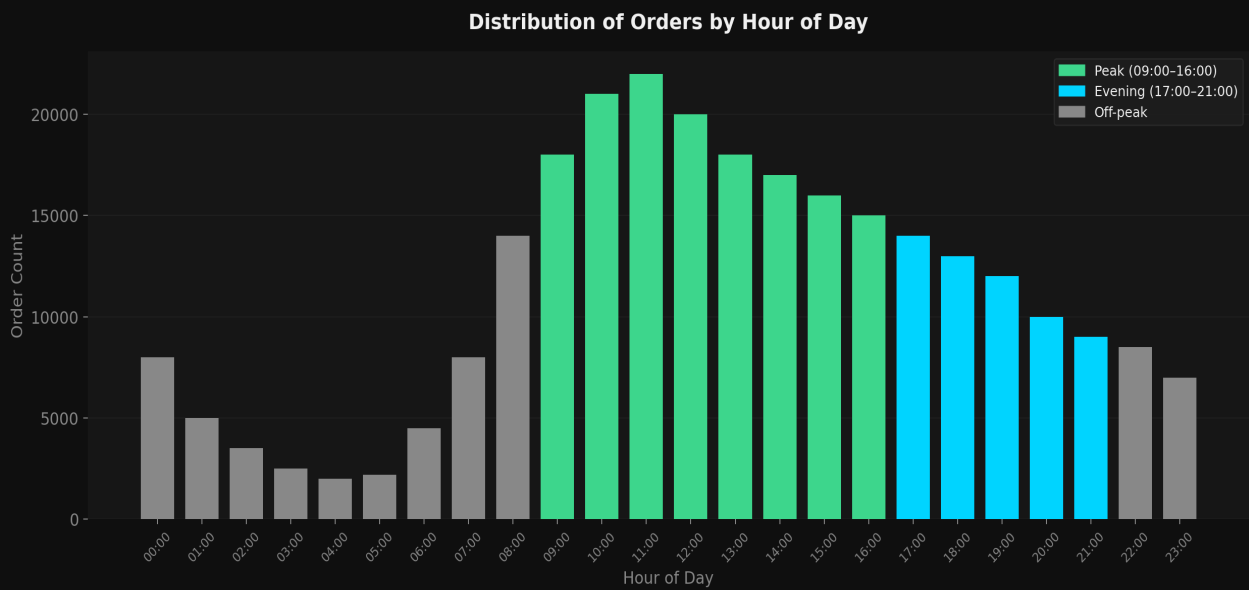
260,290 missing values in days_since_prior_order is expected — these are first-ever orders with no prior order to reference. Filling with 0 is correct. Context determines whether missing = problem.

When Do Customers Order?

Three derived variables captured order timing patterns.

Derived Timing Variables

Variable	Logic	Values
price_label	prices ≤5 → Low, 5–15 → Mid, >15 → High	Low / Mid / High
busiest_day	Top dow = Busy, bottom = Slow, rest = Average	Busy / Average / Slow
busiest_days	Top 2 days = Busiest, bottom 2 = Slowest	Busiest / Average / Slowest
busiest_period_of_day	Hours split by order count quantiles	Most orders / Average orders / Fewest orders

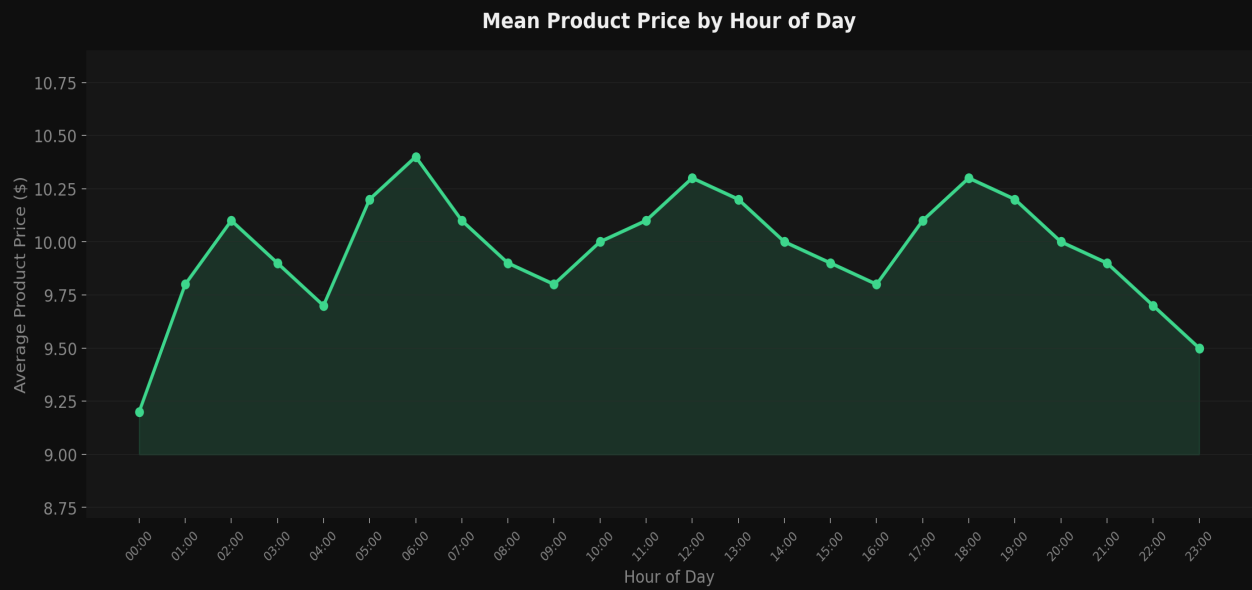


Orders peak between 9 AM and 4 PM. Instacart is primarily a daytime shopping platform.

Figure 1. Order distribution by hour. Peak activity 9 AM–4 PM. The 10:00 hour is the busiest.

KEY INSIGHT

Orders peak at 10:00 AM. Promotions timed to 8–10 AM capture customers at highest purchase intent. Late-night discount strategies are unlikely to move volume.



Average basket price is relatively stable across hours — ordering time does not drive spending.

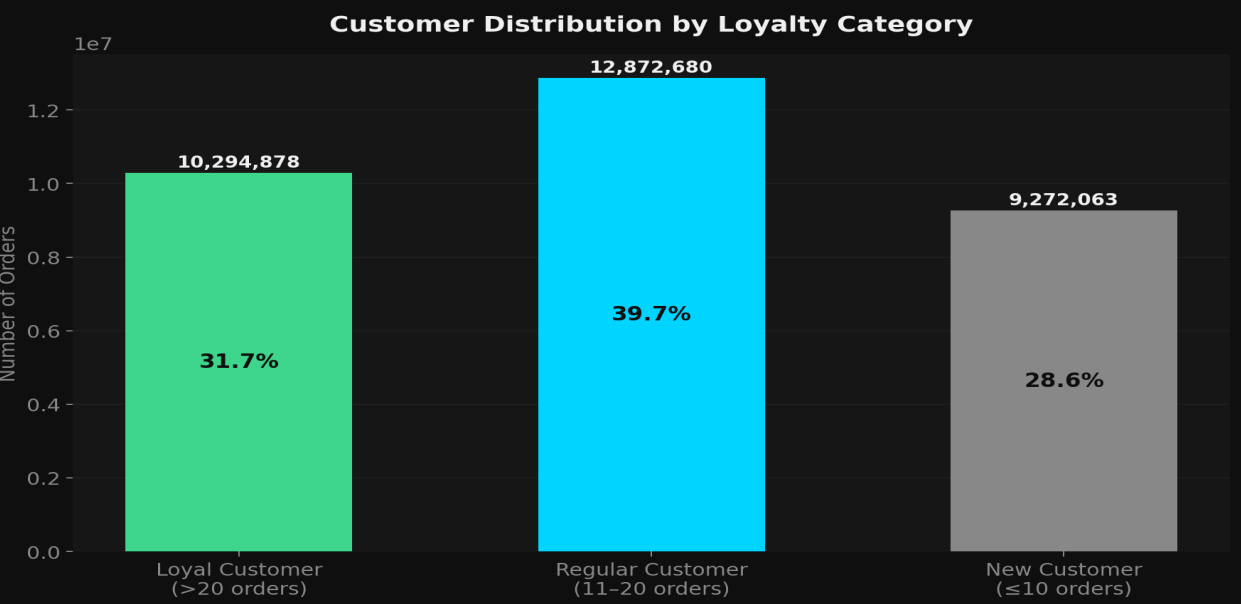
Figure 2. Mean product price by hour. Price is stable across all hours — time of day does not influence spending tier selection.

Customer Loyalty Segmentation

Loyalty flag based on mean order number per user.

Loyalty Flag Logic

Segment	Condition	Business Meaning
Loyal Customer	mean order_number > 20	Long-term, high-engagement users
Regular Customer	mean order_number 11–20	Established but not yet habitual
New Customer	mean order_number ≤ 10	Recent or infrequent users



Regular customers drive the most orders, but Loyal customers have the highest per-order value.
Figure 3. Customer distribution by loyalty category. Regular customers drive most orders by volume; Loyal customers have the highest per-order value.

KEY INSIGHT

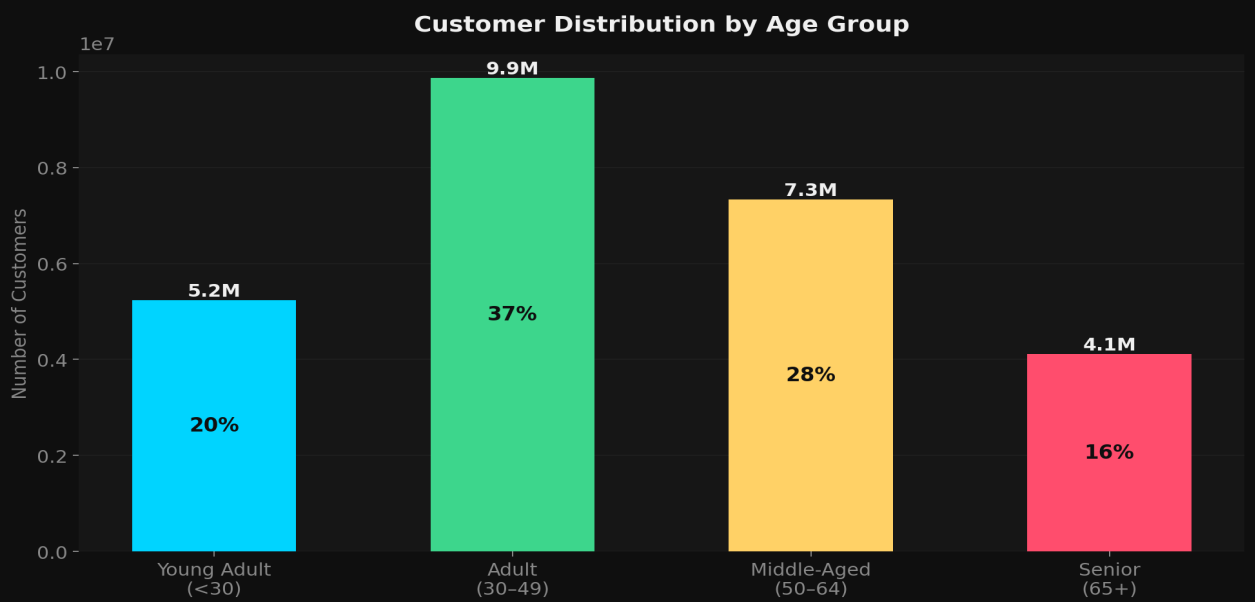
Loyal customers are not automatically high spenders. Combining loyalty and spending flags creates more actionable segments than either variable alone.

Age, Income & Regional Analysis

Four segmentation variables were created after merging 206K customer records.

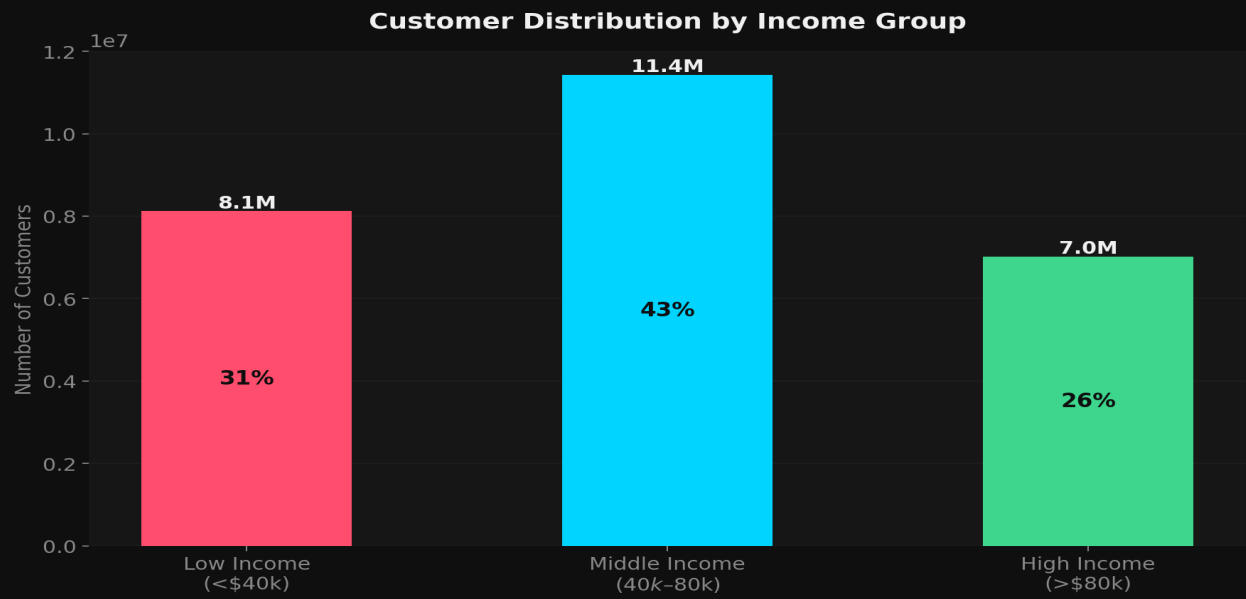
Segmentation Variables

Variable	Bins	Labels
age_group	<30 / 30–49 / 50–64 / 65+	Young adult / Adult / Middle-aged adult / Senior
income_group	<\$40k / \$40k–\$80k / >\$80k	Low income / Middle income / High income
family_status	0 / 1–2 / 3+ dependents	No dependents / Small family / Large family
order_frequency	5–9 / 10–19 / 20+ orders	Low frequency / Medium frequency / High frequency



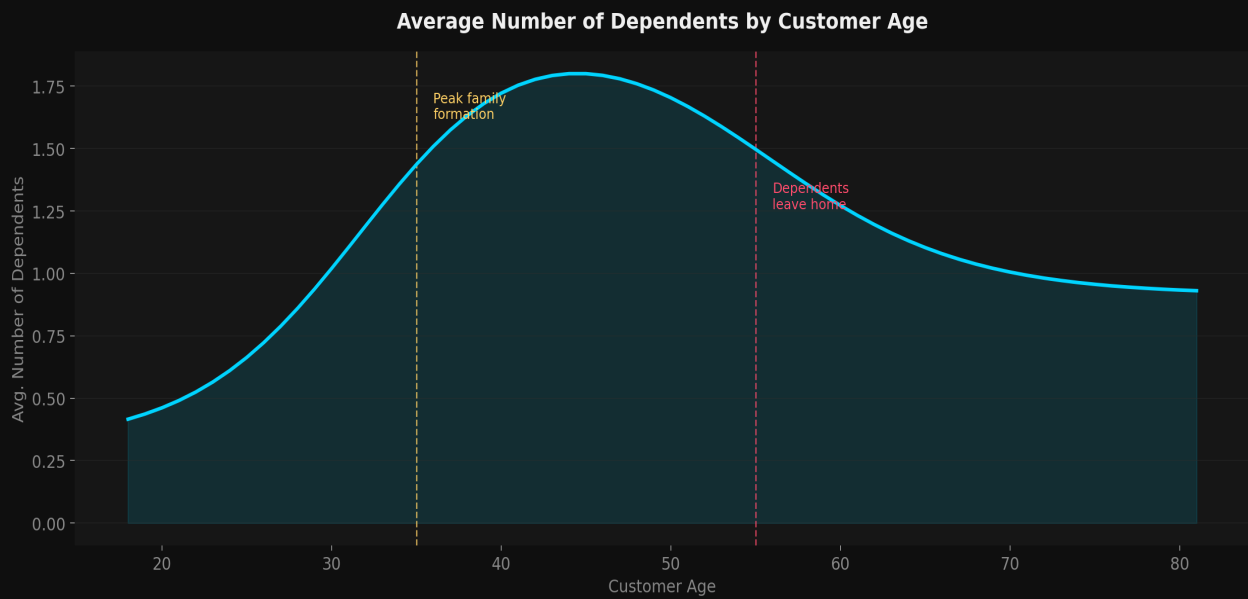
Adults (30–49) are the largest segment — the primary household grocery buyers.

Figure 4. Customer distribution by age group. Adults (30–49) are the largest segment — the core household grocery buyer demographic.



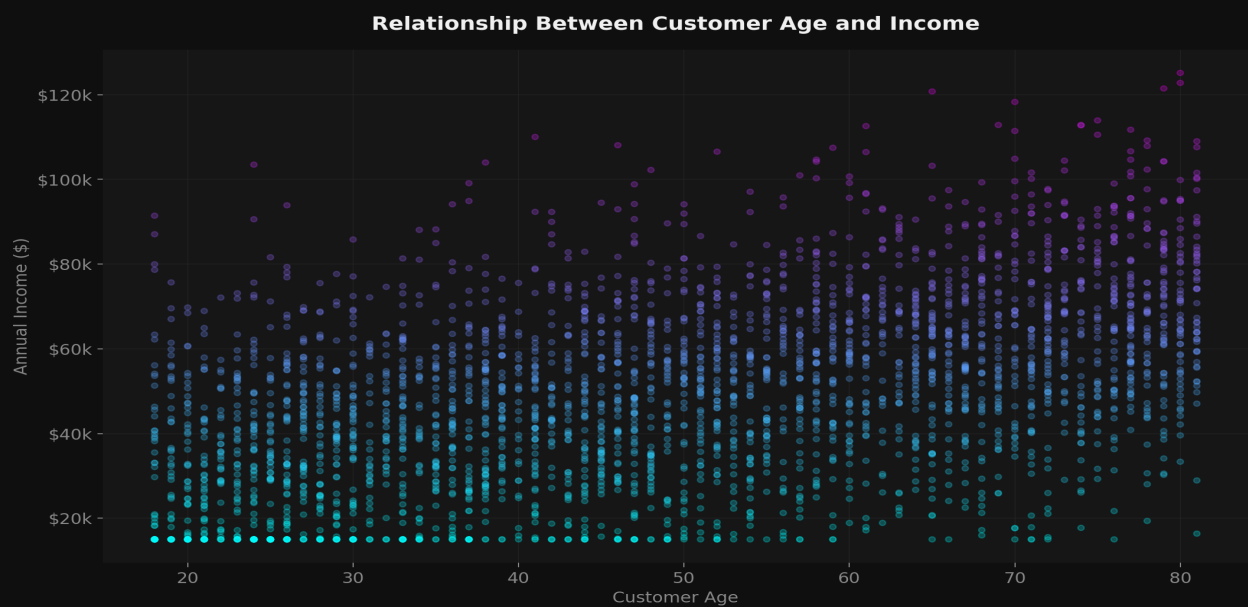
Middle-income customers dominate — representing the core Instacart grocery shopping demographic.

Figure 5. Customer distribution by income group. Middle-income customers dominate.



Life-stage pattern: dependents peak in the 30–50 age range and decline as children grow independent.

Figure 6. Average dependents by age. Classic life-stage curve — dependents peak in the 30–50 range and decline as children become independent.



Income variability increases significantly with age. No strong linear correlation — income rises but spreads widely after 30.

Figure 7. Age vs. income scatter. No strong linear correlation — income rises with age but becomes increasingly variable after 30.

KEY INSIGHT

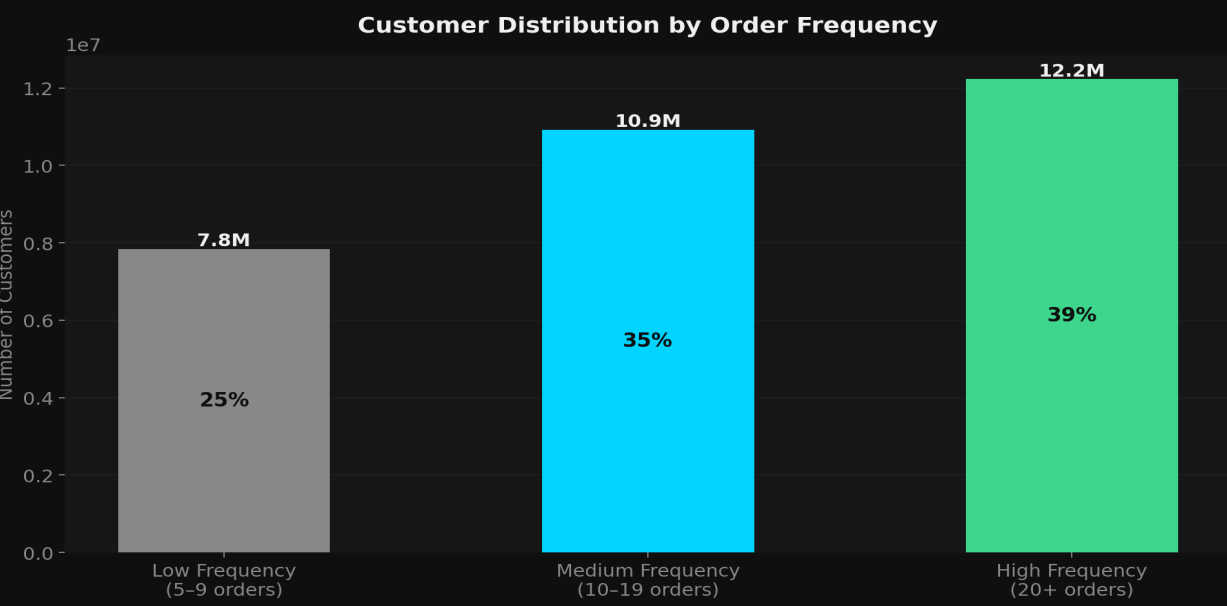
State-level analysis is too granular — 50 groups with too little data each. Mapping to 4 U.S. Census regions reduces noise while preserving meaningful geographic variation for strategy.

Building Customer Segments

Low-activity customers (fewer than 5 orders) were excluded first — removing 1.44 million rows. Each remaining customer was assigned a combined profile string.

Profile Format

Age Group | Income Group | Family Status | Order Frequency
Example: Adult | Middle income | Small family | High frequency



High-frequency customers (20+ orders) are the largest group — repeat buyers drive Instacart revenue.
Figure 8. Distribution by order frequency. High-frequency customers (20+ orders) are the largest group — repeat buyers drive Instacart revenue.

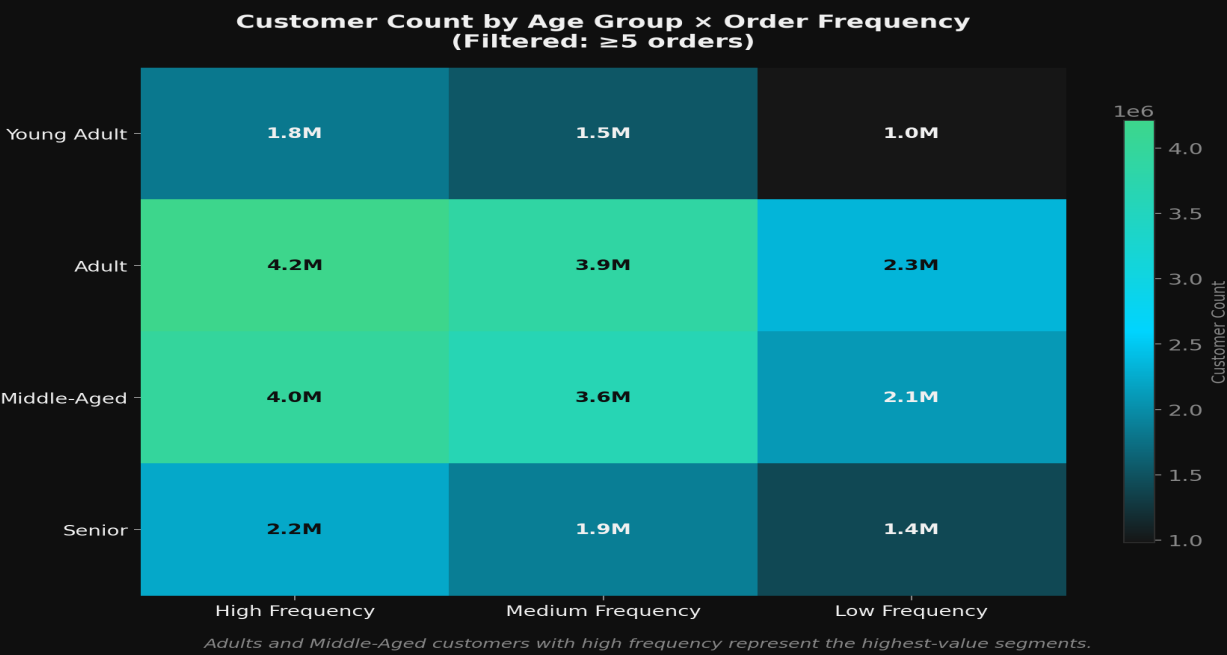


Figure 9. Customer count by age group × order frequency. Adults and Middle-Aged high-frequency customers are the highest-value segments.

**KEY
INSIGHT**

Excluding customers with fewer than 5 orders removed 1.44M rows (4.4% of data) but significantly improved signal quality. One-time users would skew segmentation results.

Business Insights & Recommendations

10 AM and 4 PM.

Instacart is a daytime platform. Promotions and staffing should cater to this behavior. Peak staffing needed around 10:00–11:00 AM.

Consistent across all hours.

Time-based pricing incentives are unlikely to change spending patterns across segments, not hours.

Regulars generate the most order volume.

A loyalty conversion strategy — moving Regular to Loyal — offers potential as Loyal customers have higher per-order value.

Young adults are the dominant segment.

Marketing should optimize for family-stage behavior: household staples, convenience items.

Young adults and Millennials are the core demographic.

Core catalog pricing should remain accessible to Middle income households. Targeted promotions for High income segments specifically.

Young adults are the largest single group.

Retention strategies (subscriptions, loyalty rewards) have a large potential impact on repeat purchases — making them higher ROI.

Limitations

- Customer data is synthetic (CareerFoundry dataset) — findings are directional, not production-validated
- No product category-level analysis — department preferences per segment were not fully explored
- No time-series component — analysis is a snapshot; seasonal trends are not captured
- Spending flag uses item price as proxy — total basket value per order would be more accurate

Technical Reference

Dependencies

Library	Purpose
pandas	Data import, wrangling, grouping, merging, export
numpy	Numeric operations and array handling
matplotlib	Bar charts, line charts, scatter plots, histograms
seaborn	Styled statistical visualizations
os	File path management across notebooks
pickle / pkl	Efficient storage of large merged dataframes between notebooks

Dataset Summary

Metric	Value
Total order rows (ords_prods_merge)	~32.4 million
Total customers	~206,000
Orders after low-activity filter	~31.0 million
Products	49,688
Departments	21
U.S. states covered	50
U.S. Census regions	4 (Northeast, Midwest, South, West)
Max customer profile combinations	108 (4 age × 3 income × 3 family × 3 frequency)

GitHub & Portfolio

Source code (all notebooks):

github.com/ageelalramadhan/Project-4-Instacart

Portfolio case study:

ageelalramadhan.github.io/instacart-case-study.html